

infection events at time s
 $i(s) = \gamma T \mathcal{V}(s)$

age α



0

$t - \alpha_2$

$t - \alpha_1$

$t - \alpha$

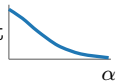
t

s

each marker starts a cohort;
 at time t , its age is $\alpha = t - s$

cohort born at $t - \alpha$

survival weight



$I_{\text{fix}}(\alpha)$: still productive

sum over ages
 $(i * I_{\text{fix}})(t)$

sum of surviving cohorts

$$\mathcal{I}(t) = \mathcal{I}_0 I_{\text{fix}}(t) + (i * I_{\text{fix}})(t)$$

weight by
 cell age

release weight



$g(\alpha) = \delta K(\alpha)$: release flux

sum over ages
 $(i * g)(t)$

release-clearance balance

$$\mathcal{V}'(t) = \mathcal{I}_0 g(t) + (i * g)(t) - c \mathcal{V}(t)$$

cohort already infected at $s = 0$: $\mathcal{I}_0$

$\mathcal{I}_0 I_{\text{fix}}(t)$

$\mathcal{I}_0 g(t)$

incidence feedback closes the loop: $i(t) = \gamma T \mathcal{V}(t)$